

Microsoft: Growth, Reinvestment and the Price of AI

A 10-Year DCF and Scenario Analysis

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NASDAQ: MSFT · UNLEVERED DISCOUNTED CASH FLOW · VALUATION DATE 8 SEPTEMBER 2026 · US\$ UNLESS STATED

PAGE 1 · INVESTMENT SNAPSHOT

A ten-year unlevered discounted cash flow model values Microsoft at \$354 per share against a market price of \$494. The gap is the subject of this analysis, not its conclusion: it quantifies the assumptions an investor must accept to justify the prevailing price.

BASE CASE DCF \$354.42 per share, FY2027E–FY2036E	REFERENCE MARKET PRICE \$493.95 close, 8 September 2026	IMPLIED DOWNSIDE 28.2% base case versus market
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Microsoft enters fiscal 2027 in the middle of a period of exceptionally elevated capital investment. Revenue compounded at 13.7% over the five years to FY2026 and operating income at 16.8%, yet free cash flow ended the period essentially where it began. The difference between those two facts is capital intensity: capital expenditure rose from 12.1% of revenue in FY2022 to 34.9% in FY2026, absorbing the cash that operating leverage generated.

This analysis values Microsoft on a ten-year unlevered discounted cash flow basis, with an explicit forecast running to FY2036 and a terminal year constructed to satisfy a formal steady-state test. The base case implies \$354.42 per share against a reference price of \$493.95. A bear case of \$242.44 and a bull case of \$455.76 bracket that estimate; the bull case — which assumes faster growth, wider margins and lower terminal capital intensity — remains 7.7% below the market.

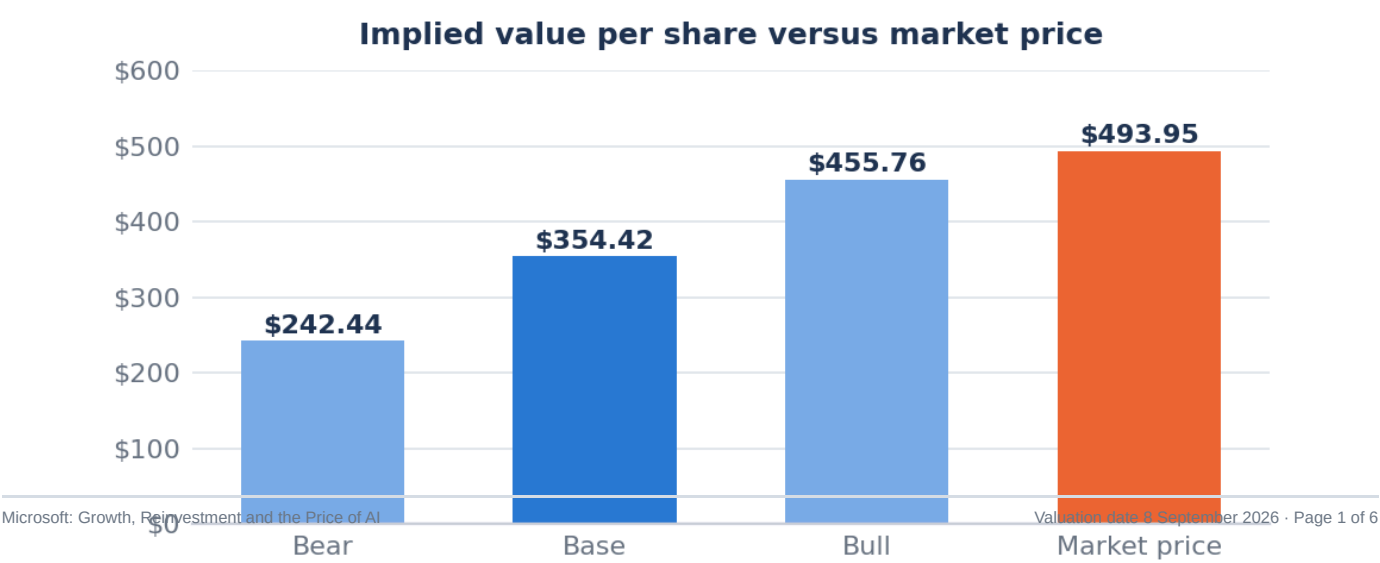
The more useful question is therefore not whether the model sits below the price, but what the price requires. Reverse-engineering the model to the market price implies either a perpetual growth rate of 5.12%, more than double the 2.50% mature-growth rate assumed here, or a weighted average cost of capital of 7.53%, corresponding to an equity risk premium of roughly 2.5% against the 4.14% applied. Neither is impossible. Both describe a substantially more demanding long-run framework, and naming them precisely is more informative than a verdict.

EXHIBIT 1.2 · VALUATION SUMMARY

MEASURE	VALUE
Base case value per share	\$354.42
Reference market price	\$493.95
Implied upside / (downside)	(28.2%)
Bear case	\$242.44
Bull case	\$455.76
Weighted average cost of capital	9.34%
Terminal growth rate	2.50%
Terminal value share of EV	60.3%
Market-implied terminal growth	5.12%
Market-implied WACC	7.53%

The valuation gap measures the assumptions embedded in the market price. It is not evidence that Microsoft is mispriced.

EXHIBIT 1.1 · SCENARIO VALUES AGAINST THE MARKET PRICE



Growth without cash conversion

Between FY2022 and FY2026 Microsoft's income statement strengthened materially while its cash generation did not. Operating income grew 86% and free cash flow grew 3%. That divergence is predominantly explained by capital expenditure.

REVENUE CAGR 13.7% FY2022A–FY2026A	OPERATING MARGIN 46.8% from 42.1% in FY2022A	FREE CASH FLOW CAGR 0.7% \$65.1bn → \$67.0bn	CAPEX / REVENUE 34.9% from 12.1% in FY2022A
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EXHIBIT 2.1 · INDEXED PERFORMANCE, FY2022A = 100

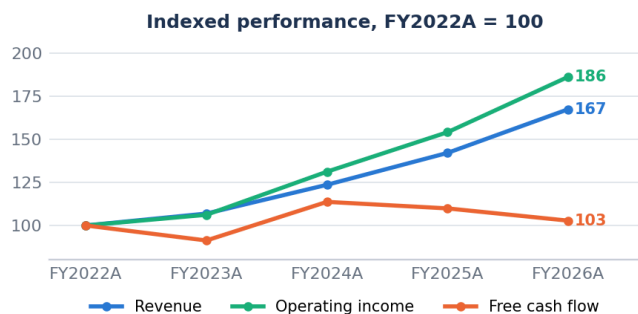
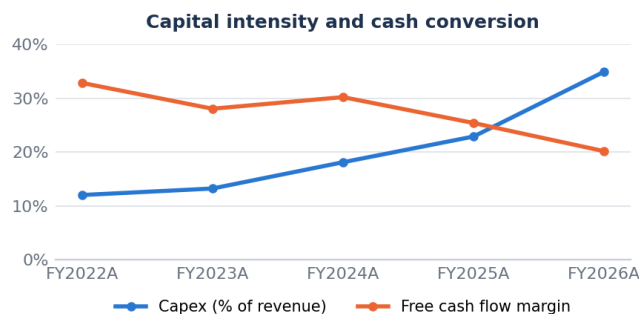


EXHIBIT 2.2 · CAPITAL INTENSITY AND CASH CONVERSION



Revenue rose from \$198.3bn in FY2022 to \$331.8bn in FY2026, a 13.7% compound rate, with growth reaccelerating to 17.8% in the final year. Operating margin expanded from 42.1% to 46.8% and net margin from 36.7% to 40.3%. On the income statement alone, the five-year record is one of consistent operating leverage.

The cash flow statement tells a different story. Free cash flow was \$65.1bn in FY2022 and \$67.0bn in FY2026 — a compound growth rate of 0.7% against revenue growth of 13.7%. Free cash flow margin fell from 32.9% to 20.2%, and cash conversion, measured as free cash flow over net income, fell from 90% to 50%. Operating cash flow more than doubled over the period; capital expenditure rose 4.9 times.

Capital intensity is the principal variable reconciling the two accounts. Capex was 12.1% of revenue in FY2022 and 34.9% in FY2026, taking net property and equipment to 0.94 times annual revenue. The balance sheet absorbed this comfortably: shareholders' equity rose from \$166.5bn to \$442.4bn, debt to equity fell from 0.30 to 0.09, and the company retained net cash of \$36.5bn. Return on equity nonetheless declined from 43.7% to 30.2%, as the retained capital base grew faster than earnings.

The FY2022–FY2026 record therefore establishes the central question the forecast must answer: whether the current investment cycle converts into future cash flow, and on what timetable.

EXHIBIT 2.3 · FIVE-YEAR SUMMARY

US\$BN UNLESS STATED	FY2022A	FY2023A	FY2024A	FY2025A	FY2026A
Revenue	198.3	211.9	245.1	281.7	331.8
Revenue growth	—	6.9%	15.7%	14.9%	17.8%
Gross margin	68.4%	68.9%	69.8%	68.8%	67.9%
Operating income	83.4	88.5	109.4	128.5	155.2
Operating margin	42.1%	41.8%	44.6%	45.6%	46.8%
Net income	72.7	72.4	88.1	101.8	133.7
Operating cash flow	89.0	87.6	118.5	136.2	182.9
Capital expenditure	(23.9)	(28.1)	(44.5)	(64.6)	(115.9)
Free cash flow	65.1	59.5	74.1	71.6	67.0
Free cash flow margin	32.9%	28.1%	30.2%	25.4%	20.2%

Capital intensity, not margin, drives the forecast

The forecast does not rely on sustained operating-margin expansion. It relies on the current investment cycle completing. Capital expenditure normalising from 34.0% of revenue to 18.6%, not margin expansion, is what drives unlevered free cash flow from \$53bn to \$252bn.

EXHIBIT 3.2 · MARGIN PROGRESSION

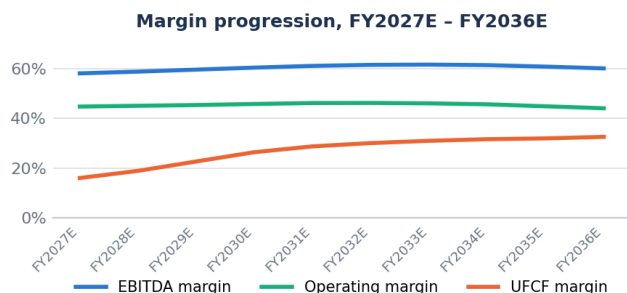
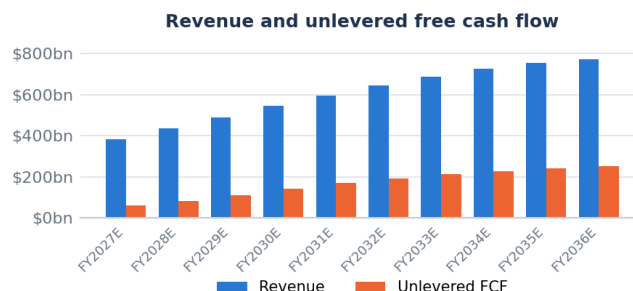


EXHIBIT 3.3 · REVENUE AND UNLEVERED FREE CASH FLOW



REVENUE FY2036E

\$772.8bn

8.8% CAGR from FY2026A

CAPEX / REVENUE

34.0 → 18.6%

\$1.45tn cumulative

PP&E DEPRECIATION

12.5 → 15.9%

% of revenue

UFCF MARGIN FY2036E

32.6%

from 16.1% in FY2027E

The explicit forecast runs ten years, from FY2027 to FY2036. Revenue growth decays from 15.5% to a terminal 2.5%, producing an 8.8% compound rate over the decade. EBITDA margin rises from 58.0% to a peak of 61.5% in FY2033 before easing to 60.0%, a range of roughly three percentage points around the FY2026 level.

Operating margin behaves differently, and the reason matters. It falls from 46.8% actual in FY2026 to 44.7% in FY2027 and does not recover to the FY2026 level at any point, ending at 44.0%. That decline is largely depreciation. PP&E depreciation rises from 12.5% of revenue to 15.9% as the current capital programme is written off, and depreciation is an operating expense. Because it is also non-cash and added back in the cash flow build, the effect on free cash flow is marginally positive

through the tax shield. Readers tracking operating margin alone would misread the forecast.

The forecast separates PP&E depreciation from intangible amortisation throughout, and only PP&E depreciation feeds the net property and equipment roll-forward. Intangible amortisation runs off from \$3.1bn to \$0.7bn on the disclosed schedule.

Capital expenditure is the dominant variable. The forecast absorbs \$1.45tn of cumulative capital expenditure while intensity declines from 34.0% of revenue to 18.6%. Net property and equipment rises from 0.94 times revenue to a peak of 1.13 times in FY2030 and then holds flat: capital intensity steps up permanently relative to the pre-investment period, but stops rising.

Terminal year construction. FY2036 capital expenditure is set so that capex less depreciation equals the terminal growth rate multiplied by opening net property and equipment — 2.71% of revenue on both sides. The final forecast year is therefore a steady state rather than a final year held flat in perpetuity, and the model carries the test as a live check.

EXHIBIT 3.1 · FORECAST DRIVERS, SELECTED YEARS

DRIVER	FY2027E	FY2029E	FY2031E	FY2033E	FY2036E
Revenue growth	15.5%	12.5%	9.5%	6.7%	2.5%
EBITDA margin	58.0%	59.5%	61.0%	61.5%	60.0%
PP&E depreciation (% revenue)	12.5%	13.8%	14.7%	15.4%	15.9%
Capital expenditure (% revenue)	34.0%	29.0%	24.0%	22.0%	18.6%
Effective tax rate	20.0%	19.0%	19.0%	19.6%	20.5%
Net PP&E / revenue	1.03x	1.12x	1.12x	1.11x	1.11x
Unlevered free cash flow (\$bn)	61.6	111.5	171.7	213.4	251.6

A transparent bridge from cash flow to value

The valuation is mechanically transparent. A ten-year explicit forecast contributes 39.7% of enterprise value and the terminal value 60.3%. The ten-year explicit horizon reduces terminal-value dependence relative to a shorter explicit forecast.

EXHIBIT 4.1 · ENTERPRISE TO EQUITY VALUE BRIDGE

US\$BN	VALUE
PV of explicit forecast (FY2027E–FY2036E)	1,034.5
PV of terminal value	1,570.4
Enterprise value	2,604.9
(+) Cash and investments	76.8
(–) Total debt	(40.3)
Equity value	2,641.5
Diluted weighted-average shares (m)	7,453
Implied value per share	\$354.42

EXHIBIT 4.2 · COST OF CAPITAL

COMPONENT	VALUE
Risk-free rate (10-year US Treasury)	4.80%
Equity beta	1.11
Equity risk premium	4.14%
Cost of equity	9.40%
Pre-tax cost of debt	5.24%
After-tax cost of debt	4.19%
Equity weight / debt weight	98.9% / 1.1%
Weighted average cost of capital	9.34%

Free cash flow is discounted at a weighted average cost of capital of 9.34%, built from a 4.80% risk-free rate, an equity beta of 1.11 and an equity risk premium of 4.14%, giving a cost of equity of 9.40%. The after-tax cost of debt is 4.19%. Capital weights use the market value of equity, with book debt applied as a proxy for the market value of debt, and because Microsoft carries \$40.3bn of debt against an equity value at the reference price of \$3,681.4bn, the debt weight is 1.1% and the discount rate is, in substance, the cost of equity.

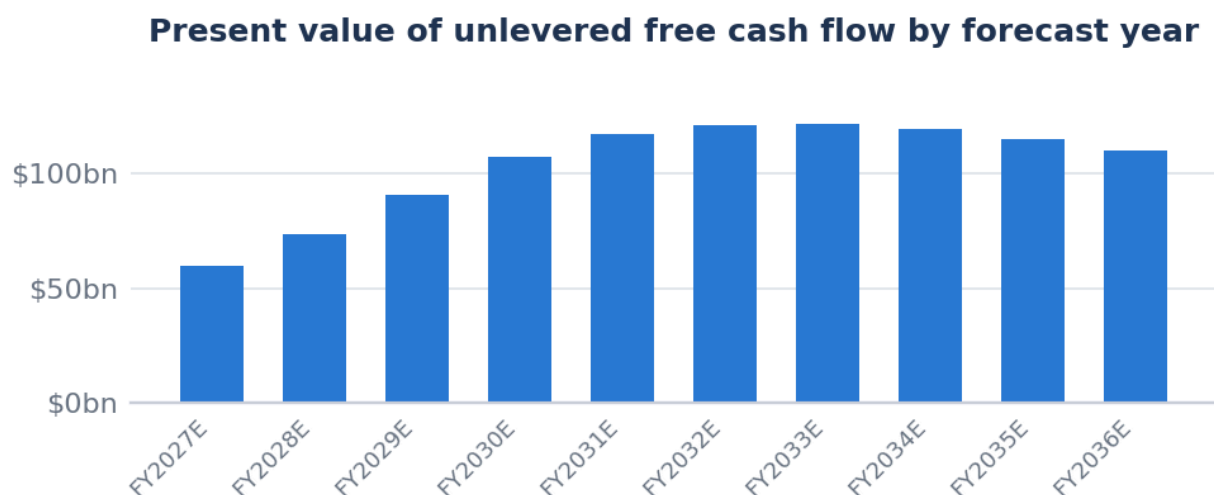
Explicit forecast cash flows are discounted on a mid-year convention, with the terminal value discounted to the FY2036 year end. The present value of the ten explicit years is \$1,034.5bn. The terminal value, calculated on a Gordon growth basis at 2.50%, has a present value of \$1,570.4bn. Enterprise value is \$2,604.9bn.

The bridge to equity is straightforward given the balance sheet: adding \$76.8bn of cash and investments and deducting \$40.3bn of debt gives an equity value of \$2,641.5bn. Divided by 7,453m diluted weighted-average shares, that is \$354.42 per share.

The terminal value carries 60.3% of enterprise value. That share is high in absolute terms and unavoidable in any discounted cash flow model of a durable business. The ten-year explicit horizon reduces terminal-value dependence relative to a shorter explicit forecast. Extending the explicit period to ten years moves value out of the terminal calculation and into forecast years that can be examined line by line — the principal reason the horizon was set at ten years.

Value split. Explicit forecast 39.7% · Terminal value 60.3%

EXHIBIT 4.3 · PRESENT VALUE OF UNLEVERED FREE CASH FLOW BY FORECAST YEAR



What the market price requires

The market price sits toward the optimistic edge of the discount-rate and terminal-growth range tested here, and above all three operating scenarios. Stating precisely what it requires is the useful output.

EXHIBIT 5.1 · SCENARIO COMPARISON

	BEAR	BASE	BULL
FY2036E revenue (\$bn)	648.8	772.8	900.4
FY2036E EBITDA margin	56.0%	60.0%	62.0%
FY2036E capex / revenue	22.5%	18.6%	16.8%
FY2036E UFCF margin	26.1%	32.6%	35.7%
Enterprise value (\$bn)	1,770.4	2,604.9	3,360.2
Implied value per share	\$242.44	\$354.42	\$455.76
Versus market price	(50.9%)	(28.2%)	(7.7%)

EXHIBIT 5.3 · BASE VERSUS MARKET-IMPLIED ASSUMPTIONS

ASSUMPTION	BASE	MARKET-IMPLIED
Terminal growth rate	2.50%	5.12%
Weighted average cost of capital	9.34%	7.53%
Implied equity risk premium	4.14%	~2.5%

The bull case remains 7.7% below the reference price. That is more informative than the base-case gap, because it indicates the market price is not reachable by adjusting operating assumptions within the range considered plausible here.

The sensitivity grid tests the two inputs to which any discounted cash flow model is most sensitive. Across seven discount rates from 8.00% to 10.50% and four terminal growth rates from 2.0% to 3.5% — twenty-eight combinations — one produces a value above the market price: an 8.00% cost of capital combined with 3.5% perpetual growth, giving \$517.97. The market price therefore sits toward the optimistic edge of the tested range rather than outside it, while remaining above all three operating scenarios.

EXHIBIT 5.2 · IMPLIED VALUE PER SHARE — WACC VERSUS TERMINAL GROWTH

	2.0%	2.5%	3.0%	3.5%
8.00%	422.78	448.74	479.89	517.97
8.50%	387.40	408.49	433.42	463.33
9.00%	357.19	374.55	394.80	418.74
9.34%	339.10	354.42	372.15	392.91
9.50%	331.10	345.55	362.22	381.67
10.00%	308.35	320.50	334.38	350.40
10.50%	288.36	298.66	310.33	323.67

Outlined cell is the base case. Shaded cell is the single combination of the twenty-eight tested that exceeds the \$493.95 market price.

Three operating scenarios were constructed on a common discount rate, so that differences between them reflect operating assumptions alone. The bear case assumes slower growth and persistently higher capital intensity — investment continuing while returns disappoint. The bull case assumes faster growth, wider margins and lower terminal capital intensity.

The reverse discounted cash flow isolates each lever. Holding operating assumptions and the discount rate constant, the market price implies a perpetual growth rate of 5.12% — materially above the 2.50% mature-growth rate assumed in the base case, and therefore a substantially more demanding long-run growth framework. Holding terminal growth at 2.50%, the market price implies a weighted average cost of capital of 7.53%, equivalent to an equity risk premium of approximately 2.5% against the 4.14% applied here.

Interpretation. These are statements about the assumptions embedded in the price, not about whether the price is correct. A discounted cash flow model that cannot reproduce a market price has demonstrated something about its own inputs; it has not established that the market is wrong. The market may be underwriting a longer period of above-trend growth than a ten-year explicit forecast accommodates, or optionality this framework does not capture. What the analysis establishes is the size and location of the difference.

Where the valuation is most exposed

The valuation's principal uncertainties are concentrated in the capital expenditure path and the depreciation schedule that follows from it. Both are identified and left visible rather than smoothed away.

The capital expenditure normalisation may not occur. The forecast assumes capital intensity falls from 34.0% of revenue to 18.6% over ten years. If demand for compute continues to compound beyond the forecast horizon, intensity may not normalise on this schedule, and the free cash flow expansion would not materialise. This is the single largest judgement in the model and the one most exposed to being wrong.

Depreciation is modelled top-down. The forecast applies depreciation as a percentage of revenue rather than from a bottom-up asset schedule, reflecting disclosed useful lives including the extension of datacenter and office building lives from 15 to 25 years effective FY2027, which materially reduces the charge on a large asset base. A bottom-up schedule built from capital expenditure by asset category and construction in progress would refine the path; those disclosures are recorded as outstanding.

Terminal value carries 60.3% of enterprise value. The ten-year explicit horizon reduces this materially relative to a five-year model, and the terminal year satisfies a formal steady-state test, but the valuation remains sensitive to terminal assumptions by construction.

The scenarios share a single discount rate. This is a deliberate convention that isolates operating assumptions from discount-rate assumptions. It does mean the bear case does not carry the higher required return a genuinely worse outcome might command.

The reverse DCF tests two levers. It varies terminal growth and the discount rate individually. It does not test combinations, nor does it capture optionality — new product categories, monetisation models not yet in the base — that a discounted cash flow framework handles poorly.

Conclusion

On the assumptions set out here, Microsoft's unlevered free cash flow supports a value of \$354.42 per share against a market price of \$493.95. The bear and bull cases span \$242.44 to \$455.76, and the bull case does not reach the market price.

The appropriate conclusion is narrower than the arithmetic invites. **This analysis does not establish that Microsoft is overvalued.** It establishes what an investor paying \$493.95 is implicitly assuming: either a perpetual growth rate more than double the mature-growth rate assumed here, or a cost of capital consistent with an equity risk premium near half the level applied, or a set of operating outcomes beyond the bull case constructed in this model. Those assumptions may prove correct. The purpose of the exercise is to make them explicit and quantified, so the decision to accept or reject them is taken deliberately rather than by default.

The record of the past five years supports the caution. Microsoft grew operating income 86% while free cash flow was flat, because capital intensity absorbed the difference. Whether the next five years reverse that pattern is the question on which the valuation turns, and it is a question about capital returns, not about growth.

BASE CASE	MARKET PRICE	MARKET-IMPLIED GROWTH	MARKET-IMPLIED WACC
\$354.42	\$493.95	5.12%	7.53%

METHODOLOGY

Ten-year explicit unlevered DCF, FY2027E–FY2036E, with a terminal year constructed to satisfy a steady-state test (capex less depreciation equals terminal growth multiplied by opening net PP&E). PP&E depreciation modelled separately from intangible amortisation; only PP&E depreciation feeds the net PP&E roll-forward. Unlevered free cash flow defined as NOPAT plus total D&A less capital expenditure plus the working capital cash impact. WACC built on the market value of equity with book debt as a proxy for market debt. Mid-year discounting on explicit cash flows; terminal value discounted to FY2036 year end; actual/365.25 day count. Sensitivity grid and reverse DCF calculated live from the forecast cash flow stream. Historical financials FY2022A–FY2026A sourced from the Microsoft FY2024 Annual Report, Microsoft FY2025 Annual Report, and Microsoft FY2026 Form 10-K and FY2026 results; historical free cash flow calculated as operating cash flow less additions to property and equipment. Forecast assumptions are the analyst's own judgement, with no consensus estimate, broker forecast or vendor projection used at any point. Underlying model is formula-driven and available on request.